CITY UNIVERSITY
MALAYSIA

FACULTY OF INFORMATION TECHNOLOGY

CITY UNIVERSITY MALAYSIA

CYBERJAYA CAMPUS

BIT1123/BISE2093/DIT1113

BIT1123 OBJECT ORIENTED PROGRAMMING

ASSIGNMENT 1 – INDIVIDUAL

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1- Introduction

Throughout this course, I completed 10 Java programming tutorials that progressively built my understanding of Object-Oriented Programming (OOP) concepts, beginning with basic Java programming and development tools and progressing to more advanced OOP concepts, file handling, and graphical user interface development.

This report reflects on my learning experience throughout the tutorials, the challenges I encountered, the improvements I made in my programming skills, and my future plan for developing my knowledge of Java and Object-Oriented Programming.

2- Knowledge Gained from Each Tutorial

Tutorial 1

Getting Started with GitHub Code spaces:

Tutorial 1 introduced me to the development environment that I would use throughout the course. I learned how to create and organize a GitHub repository and how to use GitHub Codespaces as a development environment.

I also created my first Java programs, including `HelloWorld.java` and `StudentGrade.java`. In addition, I was introduced to basic Git commands such as git init, git status, git add, git commit, and git push.

Tutorial 2

Coding in Java Programming Language:

Tutorial 2 introduced me to important Java programming concepts related to classes and objects. I learned how to create classes containing attributes such as name, age, and GPA, as well as how to use constructors to store values.

This tutorial helped me understand that classes can act as blueprints for objects. It was also my first significant step toward understanding object-oriented programming and how different files and classes can work together within a program.

Tutorials 3–4

Inheritance, Method Overriding and Polymorphism:

Tutorials 3 and 4 introduced inheritance using the `Person`, `Student`, and `Lecturer` classes. I learned how to create a parent class and extend it to create more specialized child classes.

I also learned how to use the `super` keyword to call a parent constructor and how to override methods using `Override`. Through this activity, I began to understand polymorphism and how different objects can behave differently even when they are related through a common parent class.

Tutorial 5

Encapsulation and the Student Information System:

Tutorial 5 introduced me to encapsulation and data hiding. I learned how to declare variables as `private` and use public getters and setters to control access to those variables.

This tutorial changed the way I thought about how data should be handled within a program. Initially, I found it difficult to understand why variables should not simply be declared as public. However, I learned that allowing direct access to data can lead to unintended modifications and inconsistent states.

Tutorial 6

Employee and Lecturer Inheritance:

Tutorial 6 reinforced my understanding of inheritance through an Employee and Lecturer example. I learned how a `Lecturer` class could extend an `Employee` class and inherit its attributes and functionality.

I was also introduced to the `protected` access modifier. This helped me understand that access modifiers provide different levels of access and that `protected` members can be accessed by subclasses.

Tutorial 7

Abstraction and Smart Home Appliance Management:

Tutorial 7 introduced abstraction and abstract classes. I learned how to create an abstract `Appliance` class and define abstract methods that must be implemented by its subclasses.

I learned that an abstract class cannot be instantiated directly and can instead provide a common structure or contract for its subclasses. Different appliance classes, such as air conditioner and washing machine, could then implement the required methods in their own way.

Tutorials 8–9

To-Do List Application and File I/O:

Tutorials 8 and 9 introduced file handling and combined several programming concepts into a practical application. I learned how to use `Array List` to store a dynamic collection of tasks and `Scanner` to receive input from the user.

I also learned how to save information on a text file using `BufferedWriter` and `FileWriter`, as well as how to read information from a file. In addition, I learned how to use `try-catch` blocks to handle exceptions.

This was an important stage in my learning because the application was able to save data and maintain it between different sessions. The combination of collections, user input, file handling, and exception handling made the tutorial more challenging, but it also helped me understand how multiple programming concepts can be combined to create a more useful application.

Tutorial 10

Quiz Battle with GUI:

Tutorial 10 introduced me to graphical user interface development. I learned how to use `JFrame` to create a window and components such as `JLabel` and `JButton` to create an interactive interface.

This was one of the most rewarding tutorials for me because I was able to create an application with a graphical interface rather than only using the console. Although understanding event-driven programming was challenging, seeing the final application respond to buttons and display information made the learning experience particularly engaging.

3- Challenges Encountered

Not every tutorial was difficult for me, and several of the early activities were straightforward. However, I encountered a few challenges as the tutorials became more advanced.

- One of the main technical challenges I experienced was related to GitHub. Around Tutorial 5, I experienced difficulties with performing `git push`, and the Run function in GitHub was also not working as expected for me. This affected my ability to continue my normal workflow and became one of the most frustrating parts of the tutorials.
- Another challenge was understanding some of the OOP concepts, particularly encapsulation and abstraction. These concepts required me to think differently about how classes should be designed and how data and functionality should be organized.

4- How the Challenges Were Overcome

- At the beginning of the course, I found it difficult to understand programming errors and determine what was causing them. As I completed more tutorials, I became more familiar with common errors and learned to examine the error messages more carefully. Instead of immediately considering an error as a major problem, I learned to use it as an indication of where I should investigate my code.

- For conceptual challenges, I improved my understanding by completing the practical activities and observing how changes in the code affected the program's output. Repeated practice also helped me understand concepts such as inheritance, encapsulation, and abstraction more clearly.

- The GitHub issue was more difficult because it was related to the development environment rather than simply a programming error. Discussing the problem with the lecturer and continuing to work with the available tools helped me develop patience and persistence when dealing with technical problems.

5- Improvements in Java Programming Skills

The tutorials significantly improved my Java programming skills. One of the most noticeable improvements was my ability to identify and solve programming errors.

At the beginning, I was not confident when an error appeared in my code. I often needed more time to understand what had gone wrong. With continued practice, I became more capable of reading error messages, identifying the relevant part of the code, and making appropriate corrections.

I also became more comfortable creating classes, objects, constructors, methods, and relationships between classes. Later tutorials allowed me to apply these concepts in larger applications involving collections, file handling, exception handling, and graphical user interfaces.

6- Understanding of OOP Concepts

The tutorials helped me develop a clearer understanding of the main principles of Object-Oriented Programming.

Initially, I understood programming mainly in terms of writing instructions that produce a particular output. As the tutorials progressed, I began to understand how a program can be divided into multiple classes, with each class responsible for a particular part of the application.

I learned that classes can be used as blueprints for objects and that objects can interact with one another to perform different tasks. I also developed an understanding of inheritance, which allows related classes to reuse existing functionality, and polymorphism, which allows related objects to behave differently through overridden methods.

Encapsulation taught me the importance of protecting data and controlling how information is accessed and modified. Abstraction helped me understand how common structures can be defined while allowing subclasses to provide their own implementations.

7- Future Learning Plans

In the future, I would like to improve my knowledge of Java and become more confident in developing larger applications.

- My main goal is to gain a deeper understanding of Java and apply the concepts learned in this course to more complex projects. I would also like to work on more case studies because practical problems can help me develop my ability to decide which programming concepts and structures should be used in different situations.
- In addition, I plan to continue practicing debugging and problem-solving because these skills are important for becoming a stronger programmer. I hope that with more practice I will be able to develop larger Java applications with greater confidence and independence.

8- Conclusion

Overall, the Java tutorials provided me with a gradual and practical introduction to Object-Oriented Programming. I began with basic Java programming and GitHub Codespaces and gradually progressed to classes and objects, inheritance, polymorphism, encapsulation, abstraction, file handling, exception handling, and GUI development.

Although I encountered technical and conceptual challenges during the tutorials, these experiences contributed to my development as a programmer. I improved my ability to understand programming errors and became more comfortable working with multiple classes and OOP concepts.

The most important outcome of this learning experience is that I now have a stronger foundation in Java and a clearer understanding of how Object-Oriented Programming can be used to organize and develop applications. I intend to build on this foundation by studying Java further, developing larger applications, and practicing more case studies.

9- GitHub Repository URL

My completed tutorial work and supporting materials are available in the following GitHub repository:

eltayebethar0-hub/ETHAR-object-oriented-programming-tutorials

README.md File

ETHAR-object-oriented-programming-tutorials/README.md at main · eltayebethar0-hub/ETHAR-object-oriented-programming-tutorials

THANK YOU